

Youssef Bushra Fouad

Full Stack Software Engineer

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Summary

Full-stack software engineer with experience building web applications using Next.js, TypeScript, Node.js, and MongoDB. Skilled in developing responsive user interfaces, implementing backend APIs, and delivering end-to-end features. Experienced in working with REST APIs, authentication, and modern frontend frameworks, with a focus on building reliable and user-friendly applications.

Work Experience

Web Full Stack Developer

Aug 2024 - Present

Skil-Dev | Cairo, Egypt

- Designed & developed full-stack features across backend & frontend, delivering end-to-end functionality for production systems.
- Built dynamic reusable frontend components, forms, & business logic using React & Next.js, ensuring responsive & friendly UIs.
- Integrated frontend applications with backend microservices, ensuring efficient data flow, state management, & seamless UX.
- Contributed to the design and implementation of scalable microservices-based backend systems using TypeScript, NestJS, Node.js, MongoDB, and PostgreSQL, improving system maintainability and performance.
- Designed and implemented integrations between Microsoft Dynamics 365 Finance, Odoo, and custom third-party financial systems, enabling seamless data synchronization and improving financial workflows.
- Implemented event-driven communication patterns using RabbitMQ, improving system decoupling and enabling reliable asynchronous processing across services.
- Utilized Redis for caching & performance optimization, reducing response times & improving system efficiency under high load.
- Managed database schema evolution & migrations using TypeORM, ensuring data consistency & smooth deployment processes.
- Contributed to system design decisions, including service decomposition, communication patterns, and scalability considerations.
- Improved observability by integrating monitoring and visualization tools such as Grafana, enabling better tracking of system performance and faster issue diagnosis.
- Developed RESTful APIs and implemented authentication mechanisms for secure user access.
- Increased test coverage using Jest, reducing production defects and improving overall system reliability.

Full Stack Software Engineer

Jan 2024 – Aug 2024

Block Gemini | Dubai, UAE

- Designed RESTful APIs using NestJS, TypeORM, & PostgreSQL, improving performance of complex data retrieval operations.
- Utilized GraphQL to streamline data querying and improve client-side performance, with Redis as a caching layer.
- Integrated JWT authentication for secure user access and role-based authorization in the application across microservices, enhancing security while ensuring scalability and performance.
- Documented APIs using Swagger, improving developer experience and reducing integration time.
- Led the optimization of geo-location search queries for large datasets using Elasticsearch, achieving an average query response time of 600 milliseconds (target < 1 second) while processing 10 million records.
- Contributed to frontend development tasks, including building features with Next.js 14 and styling with Tailwind CSS.
- Integrated NestJS APIs with React and Next.js frontends, ensuring seamless communication between the backend & UIs.
- Built and maintained a comprehensive logging system using Filebeat, Kibana, and Elasticsearch, enabling real-time log collection, monitoring, and visualization, while collaborating with the DevOps team to integrate these capabilities into backend systems for actionable insights into performance and system behavior.
- Collaborated with a diverse, multi-national team across Dubai, India, Romania and Egypt, enhancing communication and teamwork in a dynamic, fast-paced environment.

Projects

Car Showcase Web Application

Sep 2023

- Designed a fully responsive web application using React, TypeScript, Next.js 13 - for server-side rendering and static site generation (SSG), Tailwind CSS for styling and Git for versioning.
- Allowed users to view cars information retrieved from Rapid API endpoints, filter data using various search capabilities and view the results in chunks using Pagination to optimize performance and improve the user experience (UX) by providing faster & more responsive interface.
- Deployed the application to Vercel for seamless hosting and scalability.

Memories Management Social Media Web Application

Aug 2023

- Developed fully responsive full-stack app using MongoDB, Express.js, React.js, Node.js (MERN stack) and Redux for state management.
- Enabled smooth sharing of memories by allowing users to create, read, update, and delete (CRUD) memories, incorporating email/Google authentication, search & filtering functionalities, and pagination. Additionally, comments & recommendations (using keyword-based categorization) sections were added.
- Utilized RESTful architecture to ensure consistent and predictable interactions between the frontend and backend.

Data Dashboard Mini Cards Outlook Add-on

Jul 2023 - Aug 2023

- Coded MS Outlook Add-on using Yeoman generator, JavaScript (v1.0 & v1.1), Node.js, Express.js, and MongoDB, as a web add-on (that works perfectly for both outlook web and desktop versions), based on the most recent MS recommendations to ensure Cross-platform support and centralized deployment and distribution.
- The add-on effortlessly displays data cards within the Taskpane that contains relevant data (to the selected mail content) retrieved from the organization's DB and presents it in an organized and insightful manner, enhancing user productivity and decision-making.

Vehicle Rental System Desktop Application

Oct 2022

- Collaborated with a team in building vehicle rental desktop application in Java with Apache Derby Database based on MVC architectural pattern and delivered full software requirements specifications document (SRS) using UML diagrammed on MS Visio.
- The system was later re-designed based on the distributed system architecture using RMI for remote connections where Remote Adapter, Remote Façade & Singleton design patterns were used. Additionally, NoSQL Mongo database replaced the relational Derby database.

Accelerated Deep Neuroevolution Research — Graduation Project

Jun 2022

- Designed evolving Deep Neural Network (DNN) using hardware accelerators to solve supervised & reinforcement learning tasks.
- The hypothesized evolutionary technique was tested on image recognition task on the MNIST dataset as a supervised learning task and on simulated 3D robotic model training as a reinforcement learning task.
- The results showed an average usage of ~ 30% less training time and computing resources while maintaining an acceptable accuracy level compared to conventional DNNs results.

Internships

Dell Technologies Summer Academy

Aug 2021

Acquired technical & personal qualifications through unique learning experience covering topics such as Information and Storage management, soft skills, and hands-on experience in various technologies (AI, VR/AR, IOT & virtualization).

Advanced Computer Technology (ACT) — API Developer

Mar 2021

Implemented API endpoints in Python, Flask, and MS SQL server to connect two large and Data-intensive applications through uploading data to specific database from either flat files or other Databases using simple web interface.

Education

The British University in Egypt (BUE) – London South Bank University (LSBU)

Oct 2018 – Jul 2022

Bachelor of Computer Science, Software Engineering major - Cumulative GPA: 4.00 (Distinction with Honors)

Volunteer Experience

ACM BUE Chapter — Head of Technical Committee

Oct 2019 – Oct 2020

Led a team responsible for conducting technical sessions in competitive programming and providing technical support for the club members in areas like data structures, algorithms and time & space complexity reduction using mainly C++.

Certifications & Awards

Huawei Certified ICT Associate — Artificial Intelligence (HCIA-AI) — HUAWEI

Jun 2020 – Jul 2023

Huawei ICT Competition – Golden Medal — Cloud Computing National Finals

Feb 2021

Skills

Frontend: React, Next.js, JavaScript, TypeScript, Tailwind CSS

Backend: Node.js, Express.js, NestJS, TypeScript, TypeORM, Mongoose

Database: MS SQL Server, PostgreSQL, PL/SQL, MongoDB

API Development: REST, GraphQL, Swagger

Distributed Systems: Microservices, Event-driven architecture, RabbitMQ

Caching & Performance: Redis, Query optimization

Search & Observability: Elasticsearch, Kibana, Filebeat, Logstash, Grafana

Testing: Jest

Soft Skills: Teamwork, Presentation, Problem-solving, Critical Thinking and Innovation.

Languages: Arabic (Native), English (fluent - C1), German (A1).